Bags, Vectors & Transformers

Convolutional Neural Networks for Images · Bonus, Part 9

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Why Should Text Researchers Care About Images?

A growing share of political and social communication is visual. Some examples from published research:

Protest and collective action
detecting protest events, crowd size and violence in social media photos

Media content
who appears in news photos, how often, and in what roles

Political self-presentation
how politicians portray themselves on Instagram and in campaigns

Development and inequality
estimating poverty from satellite images of roofs, roads and night lights

The workhorse behind all of these: the convolutional neural network, the same idea as Bonus 7, with a window that slides in two directions.

An Image Is a Grid of Numbers

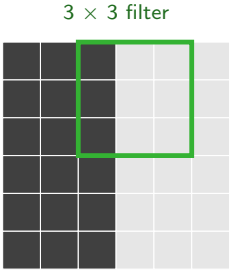
12	15	40	200	210	205
10	18	60	220	215	200
14	30	150	230	90	30
20	120	235	140	25	15
90	230	180	40	12	10
200	210	60	18	10	12

a 6×6 patch: 0 = black, 255 = white

- A grayscale image is a table with one number per **pixel**, from 0 to 255
- A color image stacks three such tables, the **channels**: red, green and blue
- A 28×28 clothing photo is 784 numbers; a phone photo, millions

Feeding pixels into logistic regression is like a bag of words for images: it has no idea which pixels are **neighbors**.

Convolution: A Filter Sliding over Pixels

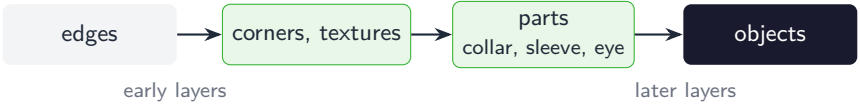


$$
* \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}
$$

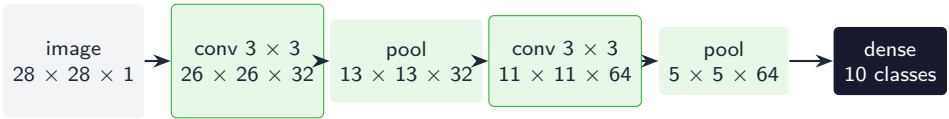
vertical-edge detector

At every position the filter multiplies the pixels under it by its weights and adds them up. This one gives a large value where **dark meets bright** from left to right: an edge.

A CNN's first layer learns dozens of filters like this **by itself**. Deeper layers combine them.



A Small CNN, Layer by Layer



Pooling keeps the maximum of every 2 × 2 block: the image gets four times smaller and a little less sensitive to exactly *where* something is.

On Fashion-MNIST, even this small CNN beats logistic regression on raw pixels by several points after a few epochs; bigger CNNs pass 90%.

One stack difference to remember: Keras stores images as height × width × channels, PyTorch as channels × height × width.

Pre-trained Models: The BERT of Images

You'll rarely train an image model from scratch. Start from one pre-trained on **ImageNet**: over a million photos in 1,000 categories. The two usage modes mirror Day 3:

1. Off the shelf

The model labels your image with ImageNet categories: *tabby cat*, *space shuttle*, *espresso*.

⚠ The categories were chosen by computer scientists. There is no "politician" or "protest" among them.

2. Image embeddings

Remove the last layer, and the model turns every image into a vector (e.g., 512 numbers). Similar images get similar vectors.

✔ Cluster your images, search them, or train a simple classifier with *your* categories on a few hundred examples.

Newer **vision-language models** such as CLIP sort images into categories you describe in words, without training, much like zero-shot text classification on Day 3. They also power the multimodal LLMs of Day 4.

Images of People: Handle with Care

Faces are personal data

Under the GDPR, photos of identifiable people are personal data, and face recognition uses biometric data, which gets extra protection. Ask your data steward first.

Models are biased

Commercial face analysis made far more errors for some groups than for others (Buolamwini & Gebru, 2018), and measures built on it inherit that.

Context and consent

A public Instagram post wasn't shared to be analyzed in a research project. Check platform terms and your ethics board.

Validate, as always

Hand-code a sample, check agreement, read the errors. Exactly as with text.

In the Notebook

- Images as arrays; edge filters and pooling, **by hand**
- A CNN for Fashion-MNIST in **Keras** and **PyTorch**, against a pixel baseline
- Looking inside: learned filters and feature maps
- Where it goes wrong: confusion matrix and misclassified images
- Pre-trained ResNet-18 and MobileNetV2: labels and image embeddings for photos
- Social science applications, ethics, and further reading

`bonus/notebooks/9_cnns_for_images_exercises.ipynb` · 6 exercises · GPU recommended